

SNMProgram It

Level 4: Step 1
HARD
Theme: Integrated Network Management
Due Date: 22/01/2022

Please take the following notes in consideration:

- In this assignment the questions marked with a (*) are practical and have to be shown to the teacher (no report is needed).
- The remaining questions are to be answered in a written report, to be delivered to the teacher.
- Whenever possible, the VMs necessary to resolve the practical questions should be created as *linked clones*.

Device Management

1. (*) Create a **Debian Linux** VM and perform the following steps:
 - (a) Install the **Net-SNMP** agent service (`snmpd` package)
 - (b) Update the **MIBs** to their latest version using the `snmp-mibs-downloader` package (non-free)
 - (c) Create a **SNMPv3** user with **authentication** and **privacy**.
2. (*) Imagine you want to provide, via SNMP, information from **IPB's** Meteorological Weather Station:
 - (a) Use the syntax from **SMIv2** to write a **MIB** file to describe the information from an hypothetical **wind-gauge** and **wind-direction** (`[0; 360]`) sensor. Declare it under the `internet/private/enterprises` node (IETF RFC1155-SMI - **.1.3.6.1.4.1**).
 - (b) On the VM created in (1) install the `build-essential` and `libsnmp-dev` packages
 - (c) Using the `libsnmp-dev` (Net-SNMP Agent) library, implement (in C/C++) the **MIB** you wrote in **2.a**).
 - (d) Install MRTG and configure it to show statistics about your **MIB**
MRTG should have a sampling period of 5 minutes

Deliverables

3. All deliverables submitted via discord#assignments-submission:
 1. PDF file with the report
 2. The file name should be `14s1h_aXXXXX_aYYYYY.pdf`. This code means:
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- Level 4
- Step 1
- Hard difficulty
- Students that participated in the group
- **Maximum:** 2 students